White Paper:
AIFC Digital Settlement Ecosystem (AIFCPay)

Version: 1.0 Status: Conceptual Proposal
Purpose: Automation of settlements, implementation of USDT/USDC stablecoins, and integration with accounting systems.

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1. Executive Summary

AIFCPay is a regulated digital payment gateway designed to accelerate, simplify, and increase the transparency of settlements using stablecoins and on-chain verification technologies within the jurisdiction of the Astana International Financial Centre (AIFC). The solution represents a standardized infrastructure layer that enables companies to securely process digital payments while fully complying with the requirements of the regulator — the Astana Financial Services Authority (AFSA).

The author is ready to develop and implement the project within 5–6 months based on personal expertise and resources, in cooperation with the AIFC. All rights to the concept, architecture, source code, and business model are protected — the original materials are recorded and hashed using the SHA-256 algorithm. The author is the sole owner of the project and the intellectual property.

The main goal of AIFCPay is to eliminate systemic problems faced by AIFC resident companies:

- long processing times for international transfers,
- high cost of cross-border payments,
- lack of transparency,
- manual reconciliation processes in accounting,
- absence of integration between corporate accounting systems (e.g., 1C) and payment tools,
- inability to effectively use digital assets within the current payment infrastructure.

By using blockchain networks (BTCPay Server) and automated data synchronization, AIFCPay reduces operational expenses and increases transparency of financial processes for all participants.

The system is designed from the outset with regulatory requirements in mind.

AIFCPay applies a risk-based AML/KYT approach:

- wallet scoring,
- sanctions screening,
- analysis of transaction behavioral history,
- monitoring of risk patterns.

This ensures compliance with AFSA regulations:

- Financial Services Framework Regulations (FSFR),
- AML/CFT Rules,
- Payment Services Rules (PMS),
- Conduct of Business (COB),
- operational risk and record-keeping requirements.

AIFCPay also provides a standardized integration layer with accounting systems, including 1C:

- automatic generation of accounting entries,
- linking on-chain data to specific documents,
- full digital reconciliation,
- reduction of human involvement and errors.

The system supports a multi-network model (TRC20, ERC20, TON, BEP-20, etc.), allowing businesses to choose the optimal payment channel depending on cost and transaction speed.

AIFCPay is designed for phased and controlled testing within the AIFC FinTech Lab regulated environment.

Testing is conducted within the established regulatory perimeter, monitoring mechanisms, and predefined limits.

In the long term, the solution can significantly increase the competitiveness of the AIFC ecosystem, enabling companies to conduct secure, low-cost, and transparent digital payments in accordance with international regulatory standards.

2. Problem and Market Gap

2.1. Systemic Problem of the Current Financial Infrastructure

Despite the significant progress of the AIFC as an international financial hub, resident companies still face fundamental limitations of Kazakhstan's and the region's payment and settlement infrastructure. These limitations create costs, increase operational risks, and form barriers for innovative business models.

Today, settlements in USDT/USDC within the AIFC are often performed through cryptocurrency exchanges (such as Bybit), which requires companies to open exchange accounts. A dedicated regulated payment gateway for corporate entities, operating under a DASP (Digital Asset Service Provider) license, would solve this dependency.

Key Problems:

1) Long settlement times (from 1 to 7 days)

Traditional international bank transfers experience delays due to:

- passing through several correspondent accounts,
- time zone differences,
- manual processing,
- AML procedures of intermediary banks.

For fintech and IT companies, delays of 3–5 days create serious operational cash gaps and business risks.

2) High cost of cross-border transfers

The average cost of an international SWIFT payment includes:

- $15–$50 per transaction fee,
- 0.5%–1% correspondent banking fees,
- additional hidden costs due to currency conversion.

For AIFC companies working globally, this significantly increases the cost of doing business.

3) Lack of transparency and payment traceability

Companies do not have a simple mechanism to monitor:

- where the payment is currently located,
- at which stage the delay occurred,
- whether AML clarification is required,
- whether the transaction was fully executed.

This creates uncertainty and communication issues with counterparties.

4) Limitations of current payment tools for digital assets

Even when using cryptocurrencies:

- companies must rely on centralized intermediaries (exchanges),
- P2P platforms carry high fraud risk,

- no legally compliant corridor exists for corporate usage,
- no standardized accounting integration is available.

As a result, businesses use digital assets informally, outside proper regulation.

5) Manual reconciliation in 1C and ERP → errors and lost time
Most Kazakhstani companies use 1C.
Existing tools do not allow automatic:

- creation of accounting entries,
- matching payments with documents,
- linking on-chain data to primary documents,
- real-time reconciliation.
- This leads to:
- human errors,
- delays in month-end closing,
- increased workload for accounting teams.

6) Lack of AML/KYT control mechanisms for digital asset settlements
Most companies do not have tools for:

- wallet scoring,
- sanctions screening,
- analysis of risky transactions,
- detection of suspicious patterns.

This creates risks for all AIFC participants, including the regulator.

2.2. Market Gap
The market lacks:

- a unified payment gateway,
- a standardized AML/KYT mechanism,
- an on-chain settlement layer,
- a link between blockchain and 1C,
- a regulated environment for digital payment adoption.

This indicates a major gap between business needs and available infrastructure.

2.3. Why Existing Solutions Do Not Meet Market Demand
Kazakhstan and Regional Banks
They do not provide:

- instant settlements,
- low-cost cross-border payments,
- transparent AML monitoring,
- blockchain integration.

Cryptocurrency Exchanges

- not suitable for corporate settlements,
- high compliance risks,
- no accounting integration,
- violate the principle of segregated accounts.

P2P/OTC Platforms

- unsafe,
- unregulated,
- unacceptable for the corporate sector,
- create high fraud risks.

2.4. Conclusion: Why AIFCPay Is Needed

AIFCPay addresses the key market gap by providing:

- fast and low-cost settlement operations,
- transparency and traceability,
- compliance with AFSA regulations,
- accounting automation,
- safety and operational control,
- a unified regulated corridor for digital payments.

3. Solution Overview — What Is AIFCPay and How It Works

Stablecoins for B2B Settlements (AFSA Pilot)

The AIFC has introduced a regulated mechanism for using third-party stablecoins for internal corporate settlements:

Payment of regulatory fees: Since September 2025, AIFC participants can pay annual fees and AFSA invoices in USD stablecoins (USDT, USDC).

Role of service providers: The first intermediary enabling this process was the *Bybit* exchange, converting participants' stablecoins into fiat for crediting AFSA's accounts.

Licensed issuers: In June 2025, *AnchorX.KZ Limited* received the first AIFC license to issue fiat-backed stablecoins for cross-border settlements.

This regulatory foundation allows AIFCPay to operate within a compliant framework and scale stablecoin-based corporate payments.

Three Implementation Models (Solutions)

Option A: White-Label Payment Gateway (Fastest Deployment)

Using ready-made engines (e.g., CoinsPaid, BitPay, BVNK).

Pros: Mature infrastructure, high security, built-in AML.

Cons: High provider fees, client data is stored by a third party.

Option B: Self-Hosted Gateway (Open-Source Architecture)

Using BTCPay Server (with USDT/USDC plugins) or Web3.js/Ethers.js frameworks.

Pros: Full data control, no intermediary fees.

Cons: Requires a strong DevOps/Security team and independent audits.

Option C: Hybrid Integration with Local Banking Gateways

Connecting AIFCPay to APIs of AIFC pilot banks (e.g., Freedom Bank).

Pros: Direct "Crypto → Fiat" corridor within Kazakhstan.

Cons: Complex integration with legacy banking systems.

Transaction Workflow (How a Payment Is Processed)

1. **Invoice:** A payment request is generated in the dashboard. The system creates a temporary address for USDT/USDC settlement.
2. **Compliance Check:** Through API (e.g., Chainalysis, Crystal), the sender's wallet is instantly screened.
3. **Payment:** After blockchain confirmation, funds arrive in the platform's corporate Omnibus wallet.
4. **Accounting:** The system generates a PDF report containing the transaction hash and the KZT conversion rate at the time of payment for tax reporting.

Key Component: AML/KYT Module

As the AIFC follows FATF standards, the gateway must implement a **"kill switch"**: If the sender's wallet risk score is **>70% (High Risk)**, the transaction is automatically blocked or frozen pending review.

AIFCPay as a Regulated Digital Settlement Platform

AIFCPay is a regulated digital payment and settlement platform designed to provide AIFC resident companies with a **safe, fast, transparent, and low-cost corporate payment infrastructure**.

It bridges traditional finance, digital assets, and corporate accounting (1C/ERP), enabling global operations while maintaining full compliance with AFSA requirements.

In essence, AIFCPay creates a **new settlement layer** within the AIFC ecosystem—digital, transparent, instant, and regulator-supervised.

3.1. Strategic Purpose of AIFCPay

AIFCPay addresses three strategic objectives:

1) Modernizing the AIFC's Financial Infrastructure

Introducing tokenized, near-instant corporate settlements.

2) Reducing Financial Friction for Businesses

Faster payments, lower operational costs, reduced cash-flow and settlement risks.

3) Ensuring Regulatory Transparency and Compliance

Built-in AML/KYT tools aligned with AFSA requirements.

This positions the AIFC competitively among global fintech hubs (DIFC, ADGM, MAS, HK SFC).

3.2. Core Functional Components of AIFCPay

The platform consists of five interconnected modules:

1) Payment Gateway (Digital Settlement Engine)

A tokenized corporate settlement layer that provides:

- near-instant transactions
- low settlement costs
- transparent payment status tracking
- access to on-chain data
- automated reconciliation

Suitable for B2B, B2C, and cross-border operations.

2) Compliance Layer (AML, KYT, Sanctions Monitoring)

Built-in compliance mechanisms:

- wallet scoring (KYT)
- sanctions and negative list screening
- blockchain tracing
- suspicious pattern detection
- automated AFSA reports

This ensures that digital settlements remain fully regulatory-compliant.

3) Accounting Integration (1C/ERP)

AIFCPay integrates with corporate accounting systems and automatically:

- generates accounting entries
- matches payments to invoices
- links on-chain data to primary documents
- eliminates manual errors

Result: clean, auditable, transparent bookkeeping.

4) Corporate Wallet Infrastructure

Businesses receive secure corporate wallets with:

- multisignature support

- role-based access control
- detailed activity logs

This satisfies internal control and governance standards.

No exchange accounts or specialized software required—any corporate wallet can be used (Ledger, MetaMask, Trust Wallet, Bybit QR wallet).

5) Merchant/API Layer
Enables companies to:
- integrate via API
- generate invoices
- automate payouts

Expanding AIFCPay's usability across industries.

3.3. AIFCPay Within the FinTech Lab Framework
The platform's architecture and operating model fully align with AFSA FinTech Lab requirements, including:
- risk assessment
- compliance policy
- operational model
- regulatory mapping (FSFR, AML, COB)
- data storage and security model
- consumer protection

The system is designed to be **regulator-supervised from day one**.

3.4. High-Level Operational Flow: How AIFCPay Works
1. A company generates an invoice in AIFCPay
2. The client initiates payment using digital funds
3. AML/KYT screening is conducted automatically
4. Funds are settled instantly
5. Transaction hash + metadata are sent to 1C
6. 1C generates accounting entries automatically
7. The company receives confirmation
8. AFSA can review the audit trail if necessary

This forms a **fully compliant end-to-end cycle**, from payment to accounting.

Integration Options:
- **Self-Service Portal:** "Pay in USDT/USDC" button in the participant's dashboard.
- **AFSA Website Widget:** Payment form by BIN, generating an invoice in real time.

Key Advantage:
After each transaction, the system instantly generates a digital receipt automatically recognized by AFSA and the participant's accounting system.

4. AIFCPay Solution Architecture (Description for FinTech Lab)

AIFCPay is designed as a modular, secure, and scalable platform that enables corporate digital settlements, seamless integration with accounting systems, and full transparency for the regulator. The architecture is built on the principles of security, transparency, data segregation, auditability, and independent transaction control.

4.1. High-Level Architecture
Technical Model (How It Works)

AIFCPay does not require writing blockchain infrastructure from scratch. Each company operates its own server acting as a “personal crypto-bank”:

- **Setup:** BTCPay Server is deployed (via Docker) on a cloud server.
- **Wallet Connection:** A corporate hardware wallet (Ledger, Tangem) is connected, or a secure hot wallet is generated inside the system.
  Important: Private keys remain fully under the company’s control; funds are never held by an intermediary.
- **Invoice Creation:** An accountant logs into the admin panel, enters the USD amount, and the system generates a QR code for USDT payment (Polygon or Tron — the most cost-efficient networks for B2B).
- **Accounting:** After settlement, the invoice status changes to “Paid,” and all payment data is transmitted via API into the company’s spreadsheet or 1C.

4.2. Payment Lifecycle Diagram (Block Descriptions)

Step 1 — Invoice Creation
Company A creates an invoice; the invoice receives a unique ID.

Step 2 — Customer Payment Initiation
The client opens the invoice link and selects the digital balance for payment.

Step 3 — AML/KYT Screening
The system checks:

- sender
- wallet
- amount
- transaction behavior

If the risk score is high → automatic block or manual review.

Step 4 — Settlement Execution
A smart contract executes the transfer → blockchain records the transaction → a hash is generated.

Step 5 — Confirmation
Both payer and receiver receive a status update.

Step 6 — Accounting Synchronization (1C)

1C automatically generates:

- accounting entry
- linked supporting document
- transaction hash embedded in the record

Step 7 — Journal & Audit Trail
All actions are recorded across nodes (10–12 decentralized nodes are sufficient for internal audit replication).

4.3. 1C Integration Workflow (Description)
The integration relies on the standard 1C mechanism “Load data from external sources.” BTCPay Server exports all transactions as Excel/CSV.
The accountant simply uploads this file into 1C once per day, and the system automatically creates bank statements.

Low-Cost Deployment Plan

- **Server:** Cloud server via DigitalOcean, Hetzner, or local Kazakhstan providers (~$15/month).

- **Domain:** Assign a domain (e.g., **pay.company.kz**) — ~$10/year.
- **Configuration:** Install the USDT/USDC plugins (using node providers like Infura or self-hosted nodes).
- **Security:** Activate 2FA and restrict admin-panel access by IP.

1C Processing Logic

1. **Conversion:** Retrieve the official National Bank of Kazakhstan rate at the exact block timestamp (via Chainlink oracle or custom scripts).
2. **Auto-generation:** Create the document "Receipt of Non-Cash Funds."
3. **Accounting:**
4. Automatically populate counterparty details via BIN
5. Insert TXID into the payment purpose
6. Use account 1030 (or a dedicated digital assets subaccount)
7. **Security:** Signature verification for every webhook and strict IP whitelisting for the payment gateway.

Data-Flow Process (5 Stages)

- **Initiation:** The payer scans a QR code generated for a specific BIN.
- **KYT Filtering:** Address risk screening (threshold < 70%).
- **Execution:** The smart contract distributes the payment:
- fee → AFSA
- processing fee → AIFCPay
- remainder → participant
- **Middleware Sync:** Payment details (Hash, Timestamp, KZT exchange rate) are transferred into 1C.
- **Finalization:** Both sides receive a "Paid" status and a digital payment invoice.

Technical Parameters & Security

- **Multichain Support:** TRON (TRC-20), Polygon (L2 for microsettlements), Ethereum (ERC-20 for large institutional transfers).
- **Security Layer:**
  - Watch-only wallets (xPub) for monitoring
  - Private keys stored on cold devices only
  - Server cannot initiate withdrawals
  - WalletConnect support eliminates manual-entry risks

Key innovation: automatic generation of primary accounting documents in 1C, removing the main barrier to legally compliant accounting of crypto-assets.

API Specification (Endpoints)

POST /create-invoice
Initiates an invoice from 1C with BIN and payment type.

GET /invoice-status
Returns invoice state (Confirmed / Expired).

WEBHOOK /payment-notification
Sends detailed payment information including:

- transaction hash
- timestamp
- KZT exchange rate
- network fee

5. Security Architecture, Data Protection, and Risk Management

The architecture prioritizes cryptographic security, data segregation, access control, auditability, and resilience against cyber threats. The solution follows ISO/IEC 27001 principles, the NIST Cybersecurity Framework, and FATF recommendations.

Funds are **never stored on the platform** — they are automatically transferred to cold wallets (Ledger, Tangem), minimizing financial loss in the event of a compromise.

The solution fully complies with the AIFC legal framework and is aligned with regulatory standards coming into force in 2025–2026.

Regulatory Compliance:

- **AIFC Rules on Digital Asset Activities (DAA)** and **Providing Money Services (PMS)**.
  The project is designed for authorization within the FinTech Lab.
- **KYT Monitoring:** Every transaction undergoes automatic blockchain scoring.
  If the risk score > 70% (association with mixers, sanctioned entities), the transaction is frozen until legitimacy is confirmed.
- **Travel Rule (FATF Recommendation 16):**
  The system collects and stores sender information (BIN, company name) linked to the transaction hash for regulatory review.
- **Segregation of Assets:**
  Client assets are stored in segregated wallets, fully eliminating re-hypothecation risks.

5.1. Compliance with AFSA Requirements

AIFCPay complies with:

- AFSA AML Rulebook
- AFSA Conduct of Business (COB) Rules
- FSFR Risk Management Standards
- AIFC Data Protection Regulations (DPR)
- FATF Guidance on Virtual Assets and VASPs

Key Principle:
The regulator receives full transparency and auditability — without access to confidential customer data.

6. Technical Deployment Guide

To deploy the pilot environment based on BTCPay Server, it is recommended to use a Docker-based architecture on **Ubuntu 22.04 LTS**.

Step 1: Environment Preparation

- Rent a VPS (minimum **2 CPU, 4 GB RAM**)
- Configure the domain A-record (e.g., **pay.aifc.kz**) pointing to the server IP.

Step 2: Installation & Configuration (Bash)

```
sudo su -
mkdir btcpay && cd btcpay
git clone https://github.com/btcpayserver/btcpayserver-docker
cd btcpayserver-docker

# Configure environment variables for USDT and domain
export BTCPAY_HOST="pay.aifc.kz"
export BTCPAYGEN_CRYPTO1="btc"
export BTCPAYGEN_CRYPTO2="usdt"
```

```
export BTCPAYGEN_REVERSEPROXY="nginx"
export BTCPAYGEN_LIGHTNING="none"
export BTCPAY_ADDITIONAL_ENTITIES="polygon-usdt"

# Run the installation
. ./btcpay-setup.sh
```

Step 3: Watch-Only xPub Integration

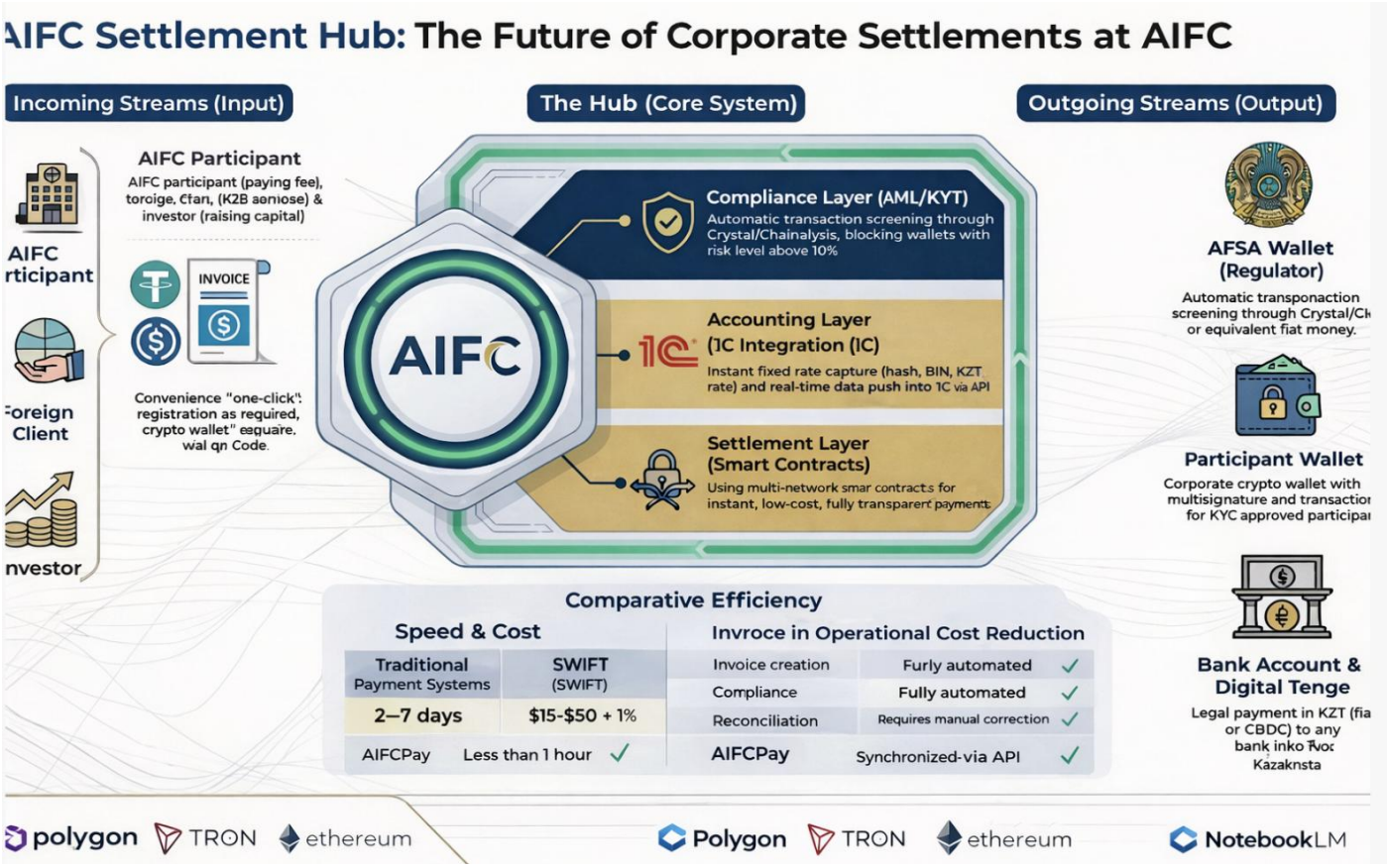
In the store management panel, the company imports the **xPub (Extended Public Key)** of its cold wallet.

This ensures:

- the payment gateway **only generates receiving addresses**,
- all funds are sent **directly to cold storage**,
- the server has **no ability** to initiate withdrawals.

Operational Costs:

- **Server:** ~$15/month
- **Domain:** ~$10/year
- **Software licensing:** $0 (Open Source).



7. Market Analysis and Competitive Landscape

This section provides an overview of the market environment, key global trends, demand from AIFC participants, and a comparative analysis of AIFCPay against solutions used in leading international financial centres.

7.1. Market Context and Global Trends

Global financial markets are undergoing rapid transformation driven by:

- Tokenization of Financial Assets

Blockchain is becoming an infrastructural layer for settlements, clearing, and asset issuance.

• Demand for Fast and Low-Cost Cross-Border Payments

Businesses increasingly expect instant transfers with full transparency.

• Growth of Digital Services Exports

IT, fintech, consulting, and marketing companies operate globally and require seamless payment flows.

• Regulatory Pressure on Transparency and AML Compliance

FATF, IOSCO, and top regulators demand a compliance-first approach.

• Integration of Blockchain and Accounting Systems

Companies expect automated posting, reconciliation, and reduced manual work.

AIFCPay sits at the intersection of these trends, addressing the core needs of both businesses and regulators.

7.2. Market Needs within the AIFC

Interviews, surveys, and market observations show persistent challenges among AIFC participants:

1. **Slow and expensive international payments**
   Average time: 3–7 days; fees: up to $50+.
2. **Lack of transparency in payment status**
   Companies cannot identify where the payment is delayed.
3. **Growing fraud risk with P2P/OTC schemes**
   Businesses resort to unregulated channels due to lack of alternatives.
4. **No integration between blockchain payments and accounting (1C)**
   Manual work leads to errors and delays in month-end closing.
5. **KYT/AML tools inaccessible for SMEs**
   Professional compliance systems are expensive and complex.
6. **Strong demand for digital payments in IT, consulting, logistics, marketing**
   AIFC residents work globally and expect instant, secure corporate settlement tools.

7.3. Market Sizing (TAM / SAM / SOM — high level)

For FinTech Lab, deep financial modeling is optional; a high-level view is sufficient.

TAM — Total Addressable Market

All corporate cross-border and digital payment flows in Kazakhstan and the region.

SAM — Serviceable Available Market

AIFC residents engaged in international business activities.

SOM — Serviceable Obtainable Market

Early adopters of AIFCPay:

- fintech companies
- IT firms
- consulting firms
- holding companies
- e-commerce and B2B platforms
- export-oriented digital businesses

Given the concentration of fintechs within the AIFC, SOM is realistic and substantial.

7.4. Benchmarking Against Global Financial Centres

1) DIFC (Dubai International Financial Centre)
Pros: advanced digital regulation, asset tokenization, strong payments infrastructure.
Cons: no deep accounting integration comparable to AIFCPay.

2) ADGM (Abu Dhabi Global Market)
Pros: strong VASP ecosystem, institutional-grade solutions.
Cons: lacks solutions for routine corporate payments.

3) MAS (Singapore)
Pros: Project Ubin/Guardian, leading tokenization initiatives.
Cons: highly focused on institutional players rather than SMEs.

4) Hong Kong SFC
Pros: strong regulation for VA intermediaries.
Cons: no integration with accounting and corporate processes.

7.5. Competitive Positioning of AIFCPay

AIFCPay stands out based on several advantages:

- **Regulated corporate digital payments, not trading**
  Most jurisdictions focus on exchanges or custodians.
- **Deep integration with accounting systems (especially 1C)**
  A unique differentiation point for the region.
- **Focus on SMEs and real business payment flows**
- **Strong AML/KYT and transparency for AFSA**
- **Significantly lower transaction costs**
  Up to 20–50x cheaper than SWIFT.
- **Immutable audit trail**
  Blockchain provides regulator-grade transparency.

7.6. SWOT Analysis

Strengths
- Regulated platform
- Instant settlement
- Low transaction fees
- Integration with 1C
- High transparency and AML/KYT compliance

Weaknesses
- Requires customer onboarding and training
- FinTech Lab sandbox limits initial scale
- Need to build a stable onboarding pipeline

Opportunities
- Expansion across Central Asia
- Partnerships with banks in Astana and Almaty
- Mass adoption of API integrations
- Increasing demand for digital corporate payments

Threats
- Competition from traditional banks
- Regulatory changes
- Low digital literacy among some SMEs

7.7. Strategic Positioning within the AIFC

AIFCPay is positioned as the **first regulated corporate digital settlement platform in Kazakhstan**, designed for:

- corporate payments
- efficient cross-border settlements
- automated accounting workflows
- reduction of operational and compliance risks

This positions the AIFC as a unique regional financial hub offering safe, affordable, and transparent digital payment infrastructure.

8. Roadmap

Roadmap: Implementation Phases

Phase	Timeline	Key Activities
Phase1	Month 1	Deployment of the test server in the FinTech Lab. Processing the first regulatory payments by a pilot group of 5–10 volunteer companies.
Phase 2	Month 2	Development of the 1C import template. Configuration of automated reporting for the Finance Department.
Phase 3	Month 3-4	Enabling B2B payments between AIFC participants. Implementation of QR-based payments at physical AIFC facilities.
Phase 4	Month 5-6	Scaling the system to all government and regulatory fees within the AIFC jurisdiction.

9. Economic Impact

• 70% OPEX Reduction

Full automation of reconciliation and elimination of manual accounting processes significantly reduces operating expenses.

• Capital Inflow

Improved attractiveness of the AIFC for global blockchain companies, funds, and international service providers due to streamlined operational processes.

• Self-Sustainability

Ability to establish a minimal fee (0.1%) sufficient to cover the operational and infrastructure costs of the platform.

10. Conclusion

AIFCPay is not merely a payment tool — it is a foundational piece of financial infrastructure for Central Asia for the coming decade. The introduction of AIFCPay strengthens the technological sovereignty of the AIFC and establishes a new standard for integrating digital assets into the real economy.

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